

David Abuín Eichinger

GRADUATE STUDENT

Research Interests

- I am interested in building autonomous machines that can adapt and coordinate with others to assist in caretaking and natural exploration.

Education

M.S.	George Mason University Computer Science	2024 - 2026
Certificate	George Mason University Computer Science, Post-baccalaureate Certificate	2017 - 2023
B.S.	North Carolina State University Microbiology Minors in Biotechnology and Genetics <i>Graduated Summa Cum Laude</i>	2009 - 2013
B.S.	United States Merchant Marine Academy Maritime Transportation	2002 - 2006
A.A.	Marion Military Institute Military Science and General Studies	2000 - 2002

Honors and Awards

Feb 2022	Joint Service Commendation Medal Exceptionally meritorious service as Strategy Plans & Strategic Sealift Planner, Knowledge Management Group, Joint Planning Support Element, Joint Enabling Capabilities Command, Norfolk, Virginia from February 2017 to December 2021
Aug 2016	Navy and Marine Corps Achievement Medal For service as Laser Weapon System Officer-in-Charge in USS Ponce from September 2015 – August 2016.
May 2009	Letter of Appreciation From Master of USNS Big Horn for professionalism and ability to perform with skill under pressure during successful combating of an out-of-control class Alpha fire on board USNS Big Horn.

Research experience

2013-2014	University of Colorado Anschutz Medical Campus Aurora, Colorado Lab of Dr. Chad Pearson • Conducted experiments to understand centriole structure and assembly in single cell organism, <i>Tetrahymena thermophila</i>.
2012-2013	North Carolina State University Raleigh, North Carolina Lab of Dr. Mike Pearson • Honors project to understand how VDJ recombination is affected in mice when subjected to DNA damage.
2012	Duke University Durham, North Carolina Summer Research Opportunity Program Lab of Dr. Daniel Lew

- Created several strains of *Saccharomyces cerevisiae* to understand rate of CDC42 polarization.

Professional Experience

2006-2023	United States Navy Lieutenant Commander • Knowledge Management Specialist • Joint Operational Planner • Strategic Sealift Officer (1665) • Information Professional Officer (1825 - Interim) • TS-SCI clearance • Honorably discharged.
2016-2019	National Geospatial Intelligence Agency Maritime Analyst , Maritime Safety Office • Writer for several world-wide regions of the publication, <i>Sailing Directions</i>
2013-2014	University of Colorado Anschutz Medical Campus Professional Research Assistant , Dr. Chad Pearson's Lab • Managed laboratory equipment and supplies. • Maintained laboratory compliance with hazardous waste and safety.
2007-2009	Military Sealift Command Third Officer , USNS Big Horn • Bridge officer on board a US government tanker that delivered fuel and supplies to ships at sea. • Trained new personnel in navigation, ship-handling, and firefighting.

Publications

2025	Chen, H., Datar, A., Xu, T., Cancelliere, F., Rangwala, H., Rao, M.B., Song, D., Eichinger, D. , and Xiao, X., "Verti-Arena: A Controllable and Standardized Indoor Testbed for Multi-Terrain Off-Road Autonomy," in <i>2025 IEEE International Symposium on Safety Security Rescue Robotics (SSRR)</i> .
2015	Galati DF, Abuin DS , Tauber GA, Pham AT, and Pearson CG. <i>Automated image analysis reveals the dynamic 3-dimensional organization of multi-ciliary arrays</i> . Biology Open, Dec 2015.

Community Service

2017-2020	Animal Welfare League of Alexandria , Alexandria, Virginia • Volunteer for cats & dogs • Foster care for cats & kittens
2014	Denver Hospice Denver, Colorado • Volunteer
2013-2014	Dumb Friends League Denver, Colorado • Dog Walker
2010	Mercy Ships Lomé, Togo • 2nd Mate, Ship's Navigator

Professional affiliations

2026- Present **American Association for the Advancement of Science (AAAS)**
2023 – Present **IEEE**
2023 – Present **Association for Computing Machinery**
2023 – 2025 **American Society for Microbiology**

Personal Interests

- Running and triathlon
- Science fiction books
- History
- Gardening
- Board games